

Elasticity Problem Set

ECON101—INTRODUCTION TO MICROECONOMICS.

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1 Warm up Exercise: Calculating Elasticities

For each demand function below, find an expression for the point-price elasticity of demand as a function of the price.

- (a) $D(p) = a - bp$
- (b) $D(p) = 60 - p$
- (c) $D(p) = \frac{40}{p^2}$. Here, the ratio $\Delta D/\Delta p$ can be approximated by $-80p^{-3}$.
- (d) $D(p) = Ap^{-b}$. The ratio $\Delta D/\Delta p$ can be approximated by $-bAp^{-b-1}$.
- (e) $D(p) = (p + 3)^{-2}$. The ratio $\Delta D/\Delta p$ can be approximated by $-2(p + 3)^{-3}$.

2 Another warm-up exercise

- (a) The computer company Pineapple has developed a new product called the *i-Widget*. The market demand for this product is $D(p) = 240 - p$.
 - i. At what price is the demand elasticity equal to zero?
 - ii. At what price is the demand infinitely elastic?
 - iii. At what price is the price-elasticity of demand equal to -1 ?
 - iv. If the *i-Widget* is priced at \$40, what is the point price elasticity of demand?
- (b) The demand for Pokemon cards is given by

$$Q_d = 500,000 - 45,000p$$

- i. At a price of \$2.5 per pack, what is the quantity demanded?
- ii. At \$5.00 per pack, what is the price elasticity of demand?

3 Elasticity, Total Revenue and Arc-Elasticity

Suppose the demand is given by $P = 22 - (1/100)Q$ and the supply is given by $P = 4 + (1/200)Q$.

- (a) Calculate the point-price elasticity of demand in equilibrium.
- (b) Calculate the point-price elasticity of supply in equilibrium.
- (c) Considering the market equilibrium point you computed above, what would happen to total revenues if the price increased? If the price decreases?
- (d) Suppose that the market was in equilibrium and then the price increased by \$2. Use the arc-elasticity formula to calculate the price elasticity of demand.

4 The market for cat treats

The demand function of cat cafes for cat treats is $q_1 = 200 - p$ and the demand function of cat owners for cat treats is $q_2(p) = 90 - 4p$.

- (a) At price p , what is the price elasticity of cat cafes for cat treats? At price p , what is the price elasticity of cat owners' demand?
- (b) At what price is the cat cafes' elasticity equal to -1 ?
- (c) At what price is the cat owners' elasticity equal to -1 ?
- (d) Graph the cat cafes' demand curve in one color, the cat owners' demand curve in another color, and the market demand curve with yet another color. Make sure to label properly your diagram.
- (e) Where on the market demand curve is the price elasticity equal to -1 . At what price will the revenue from the sale of cat treats be maximized? If the goal of the sellers is to maximize revenue, will cat treats be sold to cat cafes only, to cat owners only, or to both?

5 Income and cross-price elasticities

- (a) The income elasticity is 0.7. Income decreases by 2%, what happens to the quantity demanded?
- (b) The cross-price elasticity of demand for skies and ski poles is -1.5 . The price of a ski pole increases by 8%, what happens to the quantity demanded of skies?
- (c) The cross-price elasticity of demand for skies and skates is 1.5. The price of a skate increases by 20%, what happens to the quantity demanded for skies?